METABOLIC & GLP-1

Retatrutide

Retatrutide — 20 mg per vial — 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Retatrutide
SYNONYMS / ALIASES	LY3437943; GGG tri-agonist; GLP-1 / GIP / glucagon receptor triple agonist
CAS NUMBER	2381089-83-2
MOLECULAR FORMULA	Published structural class: lipidated 39-amino-acid peptide; exact composition described in the original Cell Metabolism (2022) reference
MOLECULAR WEIGHT	Approximately 4731 g/mol (free peptide, exclusive of counterion)
AMINO ACID SEQUENCE	39-amino-acid synthetic peptide with alpha-aminoisobutyric acid (Aib) substitutions and a C20 fatty diacid linker conjugated at a central Lys via a gamma-Glu-2x(OEG) spacer; C-terminal amide
SOURCE	Synthetic (solid-phase peptide synthesis with side-chain lipidation)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	20 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Avoid repeated freeze-thaw cycles. Like other lipidated incretin analogs, retatrutide is prone to aggregation at the air-water interface - minimize shaking and vortexing.

STRUCTURAL NOTES

Retatrutide is a 39-amino-acid synthetic peptide engineered as a triple agonist of the glucagon-like peptide-1 (GLP-1) receptor, the glucose-dependent insulinotropic polypeptide (GIP) receptor, and the glucagon receptor. The backbone uses alpha-aminoisobutyric acid substitutions to confer resistance to dipeptidyl peptidase-4 (DPP-4) cleavage, and a C20 fatty diacid is attached via a gamma-Glu-2x(OEG) linker to enable reversible albumin binding and an extended half-life.

SELECTED PEER-REVIEWED REFERENCES

1. Coskun T. et al. - *LY3437943, a novel triple glucagon, GIP, and GLP-1 receptor agonist for glycemic control and weight loss: From discovery to clinical proof of concept* - Cell Metabolism, 2022 - DOI: 10.1016/j.cmet.2022.07.013
2. Jastreboff A.M. et al. - *Triple-hormone-receptor agonist retatrutide for obesity - a Phase 2 trial* - New England Journal of Medicine, 2023 - DOI: 10.1056/NEJMoa2301972
3. Rosenstock J. et al. - *Retatrutide, a GIP, GLP-1 and glucagon receptor agonist, for people with type 2 diabetes: a randomised, double-blind, placebo and active-controlled, parallel-group, phase 2 trial* - The Lancet, 2023 - DOI: 10.1016/S0140-6736(23)01053-X

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.